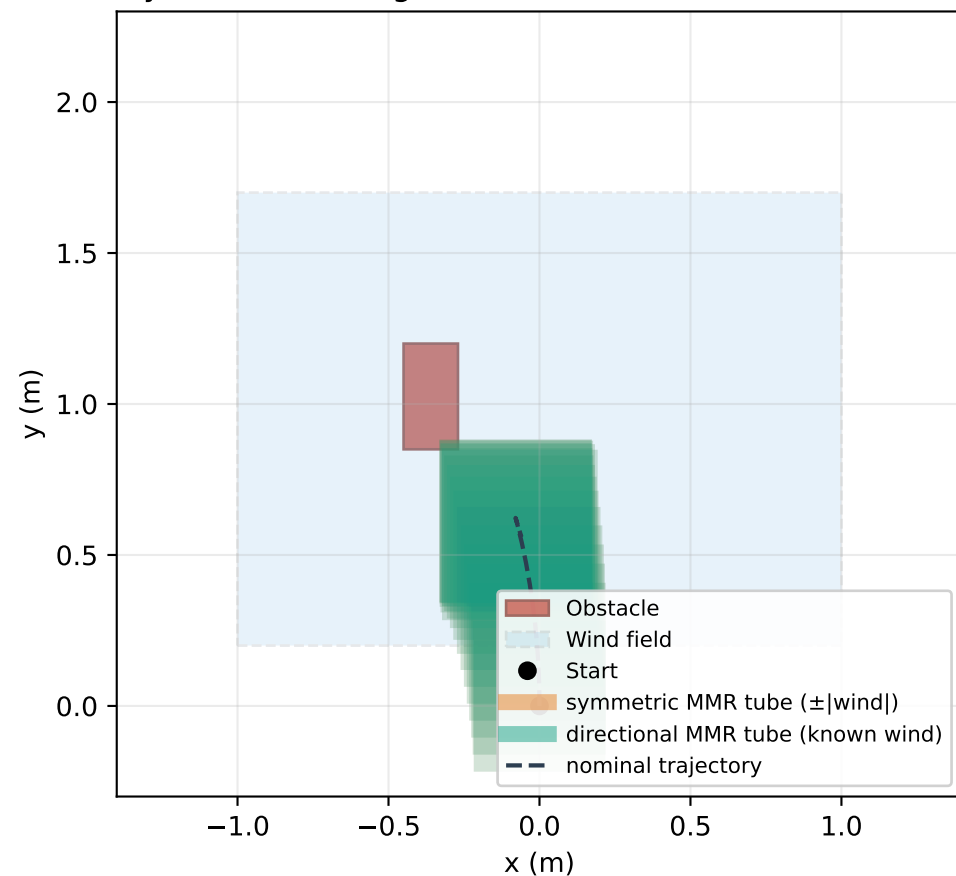


Scenario: tracking\_error — No wind. The RTD planner uses a non-inflated FRS and would declare  $k_{\text{init}}$  safe, but the MMR (immrax) tube — inflated by  $\text{init\_uncertainty}$  for tracking error — intersects the obstacle. Gradient repair on the MMR collision margin pushes  $k$  into the MMR-safe region.

Initial  $k = (0.3, 0.85)$   
symmetric (orange) vs directional (teal) MMR tubes



Final  $k$  from each method  
MMR tube envelope + nominal trajectory

